PART A — QUESTION PAPER

Model Evaluation - Comprehensive Assessment

SECTION A: Objective Type Questions (MCQs)

Q1. What is the primary purpose of model evaluation in machine learning?

A. To increase the complexity of the algorithm
B. To assess how well the model performs on unseen data
C. To reduce the size of the original dataset
D. To eliminate the need for testing data

Q2. In a standard Train-Test split, what percentage of the dataset typically comprises the training subset?

A. 10% to 20%
B. 30% to 40%
C. 50% to 60%
D. 70% to 80%

Q3. Which of the following best describes an 'overfitting' scenario in model evaluation?

A. The model performs well on training data but poorly on test data.
B. The model performs poorly on both training and test data.
C. The model performs well on both training and test data.
D. The model is too simplistic to capture underlying patterns.

Q4. In which scenario is a model said to be 'underfitting'?

A. It memorizes the training data completely.
B. It performs poorly on both training and test sets.
C. It strikes the perfect balance between complexity and generalization.
D. It performs well on the test set but poorly on the training set.

Q5. What is a False Positive (FP) in a confusion matrix?

A. When the model incorrectly predicts a negative value.
B. When the model correctly predicts a positive value.
C. When the model incorrectly predicts a positive value.
D. When the model correctly predicts a negative value.

Q6. What does 'Error' refer to in model evaluation?

A. The total number of correct predictions.
B. The total number of false positives in the model.
C. The percentage of false negatives in the model.
D. The difference between the model's prediction and the actual outcome.

Q7. When is 'Accuracy' considered an appropriate metric to use?

A. When the dataset is highly unbalanced.
B. When false negatives are extremely costly.
C. When the dataset is balanced, and both positive and negative classes are nearly equal.
D. When precision is the most important factor.

Q8. Which metric is primarily used when you want to minimize False Positives (FP)?

A. Recall
B. Precision
C. F1-Score
D. Accuracy

Q9. The F1-Score is calculated mathematically as the:

A. Sum of precision and recall.
B. Arithmetic average of precision and recall.
C. Harmonic mean of precision and recall.
D. Difference between accuracy and error.

Q10. What defines a True Negative (TN)?

A. The predicted value is positive, but the actual value is negative.
B. The predicted value matches the actual value (both are negative).
C. The predicted value is negative, but the actual value is positive.
D. The predicted value matches the actual value (both are positive).

Q11. What does a 'perfect fit' represent in model evaluation?

A. High bias and low variance.
B. Low bias and high variance.
C. The model memorizing random noise from training data.
D. The ideal balance between complexity and generalization.

Q12. In a standard confusion matrix, what do the rows typically represent?

A. The assigned values
B. The actual values of the target variable
C. The predicted values of the target variable
D. The F1 Score

Q13. Which of the following is a classic use case for a Classification model?

A. House Price prediction
B. Credit card fraud detection
C. Salary prediction
D. Temperature forecasting

Q14. What does a high Recall metric indicate about a model?

A. The model has a high rate of False Negatives.
B. The model correctly identifies most of the actual positive cases.
C. The model has zero False Positives.
D. The model is underfitting the data.

Q15. Why might a model fail to generalize to new, unseen data despite having 100% training accuracy?

A. It has underfit the data.
B. It has high bias.
C. It has overfit and memorized the training data's noise.
D. The testing subset is too large.

Q16. When is the F1 Score particularly preferred over standalone Precision or Recall?

A. When only False Positives matter.
B. When the dataset is perfectly balanced.
C. When there is a need to strike a balance between Precision and Recall.
D. When Accuracy is 100%.

Q17. What do the diagonal elements (from top-left to bottom-right) of a confusion matrix represent?

A. Only the incorrect predictions.
B. Only the False Negatives.
C. Only the correct predictions of a model.
D. Only the False Positives.

Q18. A teacher's marks prediction system predicts a student will score 75, but the actual marks obtained are 80. What is the absolute error?

A. 5
B. 10
C. 15
D. 20

Q19. Which ethical concern ensures that evaluation models treat all demographic groups equally to prevent discriminatory results?

A. Privacy
B. Fairness
C. Transparency
D. Accountability

Q20. Why can classification accuracy be misleading?

A. It cannot be mathematically calculated.
B. It is only used for regression.
C. It ignores True Positives.
D. It can be artificially high when evaluated on a highly unbalanced dataset.

SECTION B: Short Answer & Conceptual Questions

Q21. Define 'Model Evaluation' and state its primary goal in machine learning.

Q22. Explain the difference between the training subset and the testing subset.

Q23. Briefly describe what 'Underfitting' means in terms of a model's bias and variance.

Q24. Define a 'False Negative' (FN) and provide a real-world example where it might occur.

Q25. Explain why 'Accuracy' is generally considered an unreliable metric for highly unbalanced datasets.

Q26. State the core difference between the Precision and Recall metrics.

Q27. Why is the Train-Test Split technique crucial for proper model evaluation?

Q28. Briefly explain the ethical concern of 'Transparency' in AI evaluation.

Q29. In a legal AI system (innocent until proven guilty), why is Precision prioritized over Recall?

Q30. What is the purpose of the F1 Score?

Q31. Identify the specific ethical principle that ensures a developer takes responsibility for the choice of evaluation metrics and outcomes.

Q32. Explain the importance of 'Data Protection' during the evaluation process.

Q33. Define what a Confusion Matrix is and list its four primary components.

Q34. Why is Recall heavily prioritized in a medical AI model diagnosing cancer?

Q35. What does a high rate of False Positives indicate about a spam detection filter?

SECTION C: Application & Calculations

Q36. An AI model solved 85 out of 100 questions correctly on a test. Calculate the error rate of this model as a percentage.

Q37. A weather forecasting model predicts the temperature will be 32°C, but the actual recorded temperature is 28°C. Calculate the absolute error.

Q38. A credit scoring model predicts if loan applicants will default. The results show 90 correctly predicted defaults (True Positives) and 10 incorrectly predicted defaults (False Positives). Calculate the Precision of this model.

Q39. An email spam detector correctly identifies 150 spam emails (True Positives) but completely misses 50 actual spam emails, letting them into the inbox (False Negatives). Calculate the Recall of this spam detector.

Q40. A competency-based classification model outputs the following confusion matrix values: True Positives (TP) = 50, True Negatives (TN) = 40, False Positives (FP) = 10, False Negatives (FN) = 0. Calculate the total Accuracy of the model.

Q41. Using the same data from Q40 (TP=50, TN=40, FP=10, FN=0), calculate the Precision of the model.

Q42. Using the same data from Q40 (TP=50, TN=40, FP=10, FN=0), calculate the Recall of the model.

Q43. An AI evaluating social media content flags a legitimate educational video as 'inappropriate content' and removes it. In the context of a confusion matrix, classify this specific type of error (TP, TN, FP, or FN).

Q44. A financial model predicts a transaction is fraudulent. However, the transaction was actually completely legitimate. Identify this specific classification error.

Q45. If a model has a Precision of 1.0 (100%), what does this mathematically imply about its False Positives (FP)?

SECTION D: Long Answer & Multi-Part Questions

Q46. An autonomous vehicle company is evaluating its new pedestrian-detection AI system using a confusion matrix.

Define what a False Negative (FN) means in this specific self-driving scenario.

Discuss the real-world consequences of a False Negative here, and justify whether the company should prioritize maximizing Precision or maximizing Recall for this safety system.

Q47. A hospital uses machine learning to evaluate highly sensitive patient health data.

Discuss how the ethical concerns of 'Privacy' and 'Accountability' apply to this specific evaluation process.

Why must the hospital be extremely cautious about algorithmic 'Bias' when predicting patient outcomes?

Q48. A machine learning engineer notices their new classification model fits the training data perfectly (memorizing every data point) but fails completely when presented with new, unseen test data.

Identify the technical term for this phenomenon.

Describe the bias and variance characteristics of this specific model (e.g., high/low).

Explain how using the Train-Test Split methodology helped the engineer uncover this critical flaw.

Q49. Contrast the three main fitting scenarios of an AI model: Underfitting, Overfitting, and Perfect Fit. Base your comparison on how each scenario performs across training data versus testing data, and mention their ability to generalize to new data.

Q50. A data science team has generated a Confusion Matrix for a credit card fraud detection system.

Explain what the diagonal elements of this 2x2 matrix represent in terms of the model's predictions.

The team realizes they need a single evaluation metric that strikes a perfect mathematical balance between catching fraudulent transactions and avoiding false alarms. Identify the metric they should use and explain its mathematical composition.

PART B — ANSWER KEY

SECTION A - MCQ Answer Key

Q1	B	Q2	D	Q3	A	Q4	B
Q5	C	Q6	D	Q7	C	Q8	B
Q9	C	Q10	B	Q11	D	Q12	B
Q13	B	Q14	B	Q15	C	Q16	C
Q17	C	Q18	A	Q19	B	Q20	D

SECTION B & C - Short Answer and Calculation Answers

Q21: The process of applying metrics to assess an ML model's performance on unseen data.

Q22: Training (70-80%) teaches the algorithm; Testing (20-30%) evaluates it on unseen data.

Q23: It is too simplistic, failing to capture patterns. It has high bias and low variance.

Q24: The predicted value is negative, but the actual value is positive (e.g., failing to detect a disease).

Q25: The model can predict the majority class constantly and seem highly accurate while failing the minority class.

Q26: Precision focuses on exactness (reducing FP); Recall focuses on completeness (reducing FN).

Q27: It ensures the model is evaluated on unseen data, providing an unbiased estimate of real-world performance.

Q28: It ensures evaluation models do not lack clarity and clearly explain how metrics are chosen and outcomes derived.

Q29: To minimize wrongly convicting an innocent person (False Positives).

Q30: To balance Precision and Recall when the costs of FP and FN are both important.

Q31: Accountability.

Q32: It ensures individual personal data is protected against misuse or unauthorized access.

Q33: A tabular performance evaluation tool. Components: TP, TN, FP, FN.

Q34: To avoid missing any true positive cases (actual cancer patients), as False Negatives are fatal.

Q35: It incorrectly marks many legitimate emails as spam.

Q36: 15%

Q37: 4°C

Q38: 90% (or 0.9)

Q39: 75% (or 0.75)

Q40: 90% (90/100)

Q41: 83.33% (50/60)

Q42: 100% (50/50)

Q43: False Positive (FP)

Q44: False Positive (FP)

Q45: False Positives must be zero (FP = 0).

PART C — ANSWER NOTES

Q1 — Answer: B.

Explanation: Model evaluation is the process of assessing a machine learning model's performance specifically on data it has not seen before to ensure it generalizes well.

Key Concept: Purpose of Model Evaluation.

Q2 — Answer: D.

Explanation: The dataset is generally split so that 70% to 80% comprises the training subset, leaving 20% to 30% for the testing subset.

Key Concept: Train-Test Split Ratio.

Q3 — Answer: A.

Explanation: Overfitting occurs when a model learns the training data too well (memorizing noise) but struggles to generalize, leading to poor test performance.

Key Concept: Overfitting.

Q4 — Answer: B.

Explanation: An underfitting model is too simplistic and fails to capture the underlying patterns, resulting in poor performance on both training and test sets.

Key Concept: Underfitting.

Q5 — Answer: C.

Explanation: A False Positive (Type 1 error) happens when the model incorrectly predicts a positive outcome when the actual outcome is negative.

Key Concept: Confusion Matrix (False Positive).

Q6 — Answer: D.

Explanation: In model evaluation, the term Error refers strictly to the action that is inaccurate—the difference between the model's prediction and the actual outcome.

Key Concept: Error in ML.

Q7 — Answer: C.

Explanation: Accuracy is a reliable metric primarily when the dataset is balanced (equal positive and negative classes). Otherwise, it can be misleading.

Key Concept: Accuracy application.

Q8 — Answer: B.

Explanation: Precision focuses specifically on how many of the predicted positive cases were actually positive, making it the metric used to minimize false positives.

Key Concept: Precision.

Q9 — Answer: C.

Explanation: The F1-Score is mathematically defined as the harmonic mean of precision and recall, allowing it to balance both metrics equally.

Key Concept: F1-Score formulation.

Q10 — Answer: B.

Explanation: A True Negative occurs when the model accurately predicts a negative value, and the actual real-world value is also negative.

Key Concept: Confusion Matrix (True Negative).

Q11 — Answer: D.

Explanation: A perfect fit effectively balances complexity and generalization, allowing it to perform well on training data and generalize properly to new data.

Key Concept: Perfect Fit / Bias-Variance Tradeoff.

Q12 — Answer: B.

Explanation: In a standard confusion matrix layout, the rows traditionally represent the actual values (true outcomes), while the columns represent the predicted values.

Key Concept: Confusion Matrix Structure.

Q13 — Answer: B.

Explanation: Credit card fraud detection is a classification problem because it categorizes data into discrete classes (Fraud vs. Not Fraud), unlike regression which predicts continuous numbers (like prices/salaries).

Key Concept: Classification Use Cases.

Q14 — Answer: B.

Explanation: High recall means the model is highly capable of identifying actual positive instances in the dataset, effectively minimizing False Negatives.

Key Concept: Recall Metric.

Q15 — Answer: C.

Explanation: 100% training accuracy usually indicates Overfitting. The model has memorized the exact training data (and its noise) but lacks the flexibility to handle new data.

Key Concept: Overfitting consequences.

Q16 — Answer: C.

Explanation: F1 Score is the harmonic mean of Precision and Recall. It is preferred when you need a single metric that balances the costs of both False Positives and False Negatives.

Key Concept: F1 Score Utility.

Q17 — Answer: C.

Explanation: The diagonal elements from top-left to bottom-right of a confusion matrix contain the True Positives and True Negatives—the entirely correct predictions.

Key Concept: Confusion Matrix Elements.

Q18 — Answer: A.

Explanation: Absolute error is the numerical difference between the predicted value and the actual value. |75 – 80| = 5.

Key Concept: Absolute Error Calculation.

Q19 — Answer: B.

Explanation: Fairness is the ethical principle ensuring that models treat all demographic, socio-economic, and gender groups equally, avoiding discriminatory biases.

Key Concept: Ethical Concerns (Fairness).

Q20 — Answer: D.

Explanation: On an unbalanced dataset (e.g., 99% healthy, 1% sick), a model can predict 'healthy' every time and achieve 99% accuracy, while entirely failing its true objective.

Key Concept: Limitations of Accuracy.

Q36 — Answer:

Explanation: 100 total – 85 correct = 15 incorrect. Error rate = (15 / 100) × 100 = 15%.

Key Concept: Error Rate Calculation.

Q37 — Answer:

Explanation: Absolute Error = |Predicted – Actual| = |32 – 28| = 4.

Key Concept: Absolute Error Calculation.

Q38 — Answer:

Explanation: Precision = TP / (TP + FP) = 90 / (90 + 10) = 90/100 = 90%.

Key Concept: Precision Calculation.

Q39 — Answer:

Explanation: Recall = TP / (TP + FN) = 150 / (150 + 50) = 150/200 = 75%.

Key Concept: Recall Calculation.

Q40 — Answer:

Explanation: Total cases = 50+40+10+0 = 100. Accuracy = (TP + TN) / Total = (50 + 40) / 100 = 90%.

Key Concept: Accuracy Calculation.

Q41 — Answer:

Explanation: Precision = TP / (TP + FP) = 50 / (50 + 10) = 50/60 = 83.33%.

Key Concept: Precision Calculation.

Q42 — Answer:

Explanation: Recall = TP / (TP + FN) = 50 / (50 + 0) = 50/50 = 100%.

Key Concept: Recall Calculation.

Q43 — Answer:

Explanation: The actual state is legitimate (Negative for inappropriate), but the model predicted it as inappropriate (Positive).

Key Concept: Identifying Confusion Matrix Outcomes.

Q44 — Answer:

Explanation: The model flagged it as Fraud (Positive), but the reality was Legitimate (Negative).

Key Concept: Identifying Confusion Matrix Outcomes.

Q45 — Answer:

Explanation: Since Precision = TP / (TP + FP), the only way for the result to equal 1 (100%) is if FP equals 0.

Key Concept: Mathematical implications of Precision.

PART D — QUICK REVISION NOTES

1. Model Evaluation Fundamentals

• Purpose: To assess a machine learning model's performance on unseen data to ensure it generalizes well to the real world.

• Train-Test Split: Divides the dataset to prevent biased evaluations.

• Training Subset (70%-80%): Used to teach the algorithm patterns.

• Testing Subset (20%-30%): Used to evaluate the model on unseen data.

2. Model Fitting Scenarios

• Underfitting: The model is too simple. Performs poorly on both training and testing data. (High Bias, Low Variance).

• Overfitting: The model memorizes training data (including random noise) but fails to generalize. Performs well on training, but poorly on testing. (Low Bias, High Variance).

• Perfect Fit: The ideal balance between complexity and generalization. Performs well on both subsets.

3. The Confusion Matrix

• A 2×2 tabular tool used to evaluate classification models.

• True Positive (TP): Model predicts positive, actual is positive (Correct).

• True Negative (TN): Model predicts negative, actual is negative (Correct).

• False Positive (FP / Type 1 Error): Model predicts positive, actual is negative (False Alarm).

• False Negative (FN / Type 2 Error): Model predicts negative, actual is positive (Missed Detection).

• Note: The diagonal elements (TP and TN) represent the entirely correct predictions.

4. Evaluation Metrics & Formulas

• Error: The difference between prediction and actual outcome.

• Accuracy: (TP + TN) / (TP + TN + FP + FN). The proportion of total correct predictions. Warning: Highly misleading if used on unbalanced datasets.

• Precision: TP / (TP + FP). Focuses on exactness. Used when minimizing False Positives is critical (e.g., Spam Filters, Legal Systems).

• Recall: TP / (TP + FN). Focuses on completeness. Used when minimizing False Negatives is critical (e.g., Cancer Diagnosis, Fraud Detection).

• F1-Score: The harmonic mean of Precision and Recall. Used to strike a balance when both FP and FN costs are important.

5. Ethical Concerns in Evaluation

• Bias & Fairness: Models must treat all demographic groups equally to avoid discrimination.

• Transparency: Models shouldn't be 'black boxes'; evaluation metrics and results must be clearly explained.

• Accountability: Developers must take responsibility for ethical decision-making and model errors.

• Privacy & Data Protection: Real-world testing data (like medical or financial records) must be secured against misuse.